Anchor, Don't Name: What an Ontology Actually Contributes to LLM-Built Knowledge Graphs

Extended Abstract Track Submissions

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Abstract

Enterprises increasingly build knowledge graphs from their documents with LLM extractors, and in regulated domains such as finance the answers drawn from those graphs must be verifiable, not merely plausible. The standard prescription is to guide extraction with a domain ontology such as FIBO. Yet whether the ontology improves the graph — and whether the graph then improves answering — has rarely been measured with the schema as the only variable and memorization controlled. We measure both, in a pre-registered study on the FinDER financial-QA corpus: the same filings are extracted under five schema conditions, from no schema to full FIBO, by three LLMs, with the two decisive conditions replicated on 420 stratified cases. Three results follow. First, the ontology does not improve extraction: it lowers cross-model agreement on entity names and does not raise coverage of answer-bearing facts; its measurable benefit is that schema violations become detectable. Second, comparing facts by entity name — standard alignment practice — hides most cross-model disagreements: aligning each extracted value by its exact position in the source document instead yields 1.7–2.8× as many comparable facts and exposes that a quarter of anchored values carry unit-scale errors. Third, standard QA scoring favors source text over the serialized graph, but cannot distinguish answers grounded in the supplied evidence from answers recalled from pretraining: separating memorized answers, honest refusals, and grounded answers reverses the comparison on two of three models, and adding provenance pointers alone raises accuracy. We conclude that facts should be aligned by provenance rather than by name, that ontologies are better deployed as validators than as extraction guides, and that evaluations of retrieval structure must condition each verdict on what the supplied evidence could support. Eleven registered hypotheses failed and are reported as findings.

1 Introduction

Enterprises that build knowledge graphs from documents with LLMs face a design decision early: whether to guide extraction with a domain ontology. The conventional answer is yes — a schema should make extraction more consistent, more complete, and easier to query. Financial services is the domain where this belief is most institutionalized: FIBO [1], an OWL ontology of financial concepts, exists precisely to standardize how institutions name and relate the entities in their filings. Yet existing comparisons vary the schema together with prompts, models, or chunking [2–4], so the guidance's own contribution has not been isolated.

This paper measures that belief end to end on FinDER [5], a financial question-answering dataset built from SEC filings, and reports three findings that revise it. One number previews all three. We ran three LLM extractors over the same filings and compared the numeric facts they produced whenever two extractors drew a number from the same position in the source document. Their extracted values disagreed 1,482 times — and **in all 1,482 cases the disagreement is the same kind**: the digits match, and the extractors differ only on whether a table header such as "in thousands" applies. Matching facts by entity name hides every one of these conflicts; matching them by source position exposes them; and no ontology in the prompt prevents them.

Three findings, each stated with its paired confidence interval in Section 3, compose into one conclusion. The first removes the assumed mechanism: the ontology does not standardize what extractors say — it lowers cross-model agreement on entity names and covers fewer of the numbers the gold answers require, and what survives is a validation surface. The second identifies the failure that actually needs catching — scale conflicts — and shows that the name-based alignment the field builds on cannot see it, while source position can. The third shows the same blindness repeated at evaluation time: once memorization and honest abstention are separated from grounded correctness, whether the graph helps answering becomes a measurable property of the question stratum and the model. One primitive survives scrutiny in all three findings: the position in the source document from which each value was extracted. We therefore conclude that *the value of structure is not that it yields better answers but that it yields verifiable ones — and verification requires provenance, not names*. Every hypothesis above was pre-registered together with the outcome that would falsify it, before the corresponding experiment ran (Section 2); eleven were falsified and are reported as findings, and exploratory results are labelled as such and replicated under new pre-registrations. Section 2 fixes the design; Section 3 states the findings; Section 4 turns them into deployment rules; the appendix holds full results and every registration.

2 Study design

Every experiment follows one pipeline. A FinDER case supplies a question, a gold answer, and the filing passages that support it. Three LLMs each extract a knowledge graph from the same passages under one schema condition, and the three graphs are aligned fact by fact (Sections 3.1–2). The same models then answer the question under five evidence conditions — nothing, the gold passages, or a serialized graph — and every answer is scored against the gold answer and against what its evidence could support (Section 3.3).

Three samples cover FinDER's annotation grid (balanced 280; arithmetic 140; a 204-case census of the reasoning stratum). The five schema conditions each isolate one belief about what guidance contributes: a schema-free floor (how much agreement arises with no guidance at all), a hand-written 20-class schema (a small bespoke schema suffices), corpus-scoped FIBO — the 70 classes ranked by how often the corpus itself mentions them (the standard ontology, as the corpus uses it) — and FIBO plus synonyms, respectively FIBO plus its class hierarchy (does richer vocabulary, or hierarchical structure, improve extraction). The conditions nest, so each contrast is attributable to a single addition (Table 3), and everything except the schema is byte-identical and hash-recorded.

Two instruments do the work the field usually delegates to names and gold answers. **Provenance-anchored alignment**: every extracted value is located at the unique numeric token it came from in the source, so facts align across extractors by source position, not by what each model chose to call them.

Evidence-conditional evaluation: every answer verdict is conditioned on whether the served evidence contains the gold numbers, mechanically separating grounded correctness from memorization and honest abstention from failure.

Both instruments are deterministic. An LLM-judge panel serves only as a secondary check: judges never grade their own model's answers, and a judge's verdicts count only after it passes four calibration controls whose correct outcomes are known by construction.

Every uncertainty interval in this paper is a 95% confidence interval obtained by nonparametric bootstrap over *cases* (5,000 resamples, fixed seed), and every comparison between conditions is computed as a paired within-case difference before resampling, so per-case difficulty cancels; we call a difference *separated* when its interval excludes zero. In the appendix we abbreviate a difference and its interval as $d\ [a, b]$; the main text spells every interval out. Hypotheses were registered with their disconfirming outcomes before each run, and the registrations are committed files in the supplementary material. Full design, prompts, and protocol: Appendix B.

3 Findings

3.1 The ontology provides detectability, not better extraction

We measure agreement as the share of a case's facts for which two or more extractors produce the same normalized entity name, and content as the share of the gold answer's numbers present in the graph. With everything but the schema fixed, ontology guidance *lowers* agreement: the schema-free condition exceeds FIBO by 0.069, with a 95% confidence interval of 0.046 to 0.091 over the same 280 cases. It also *reduces* content coverage: the FIBO graph holds 0.036 less of the gold answers' numbers (interval 0.008 to 0.067) — the registered content-advantage hypothesis, reversed at scale. What survives is a validation surface: across the 280 cases, 529–1,038 schema violations per ontology condition become mechanically checkable where the schema-free graph offers almost nothing to check. The synonym condition never mattered because it could not: only 0.5% of the corpus's questions need vocabulary mapping, and the abbreviations they use are analyst shorthand, not FIBO's. The full five-condition dose–response, with intervals, is Table 4 in Appendix C.

3.2 Extractors never disagree about digits — only about scale

Aligning facts by anchor instead of name yields 1.7–$2.8\times$ as many comparable cross-model facts, and 80–84% of the disagreements it exposes are invisible to any name key (registered replications on both conditions at 280 cases). Auditing those disagreements produced this study's sharpest fact: all 1,482 of them are unit-scale splits. A representative case, verbatim from the artifacts: at one source token, one extractor stored the fact *"revenue software licenses"* $=$ $945{,}797$ while another stored *"software licenses revenue fy2021"* $=$ $945{,}797{,}000$ — same digits, a 10^3 split from a table's "in thousands" header, and two names no string matcher would pair — invisible to name-keyed pipelines, surfacing only as an unexplained thousand-fold error downstream. Overall, 26.7% of anchored values match their source only after rescaling by 10^3 or 10^6. Two deployment consequences follow: name-keyed entity resolution silently discards the verification signal a second extractor was supposed to buy, and cross-extractor verification has a hard ceiling regardless of key — 87% of facts are produced by a single extractor only, a share that *rises* with sample size — while serving a fact only when two extractors agree discards four correct answers for every wrong answer it blocks (registered, disconfirmed; Appendix D).

3.3 Whether the graph helps answering depends on the question stratum and the model

Under standard scoring, gold passages beat the serialized graph. But closed-book performance spans 0.06–0.31 across models on the same public filings, so a correct answer is weak evidence that the model used the material it was given. Decomposing every verdict by whether the served evidence could support it (Table 2) reverses the reading on the prose stratum — the graphs' grounded-correct yield matches or beats passages on two of three models, passages yield 2–$4\times$ as many memorized-but-unsupported answers, and the third model's deficit is over-refusal, not misunderstanding. On the arithmetic stratum the reversal does *not* replicate: computing a value needs co-located operands that extraction fragments, and the original graph-favorable hypothesis closed negative on its own chosen ground. Attaching provenance pointers — position references only, no text — raised accuracy on two of three models and repaired the over-refusing model's deficit (Appendix E).

The third stratum completes the argument and adds the study's only graph-over-text separation. On the 204-case census of questions FinDER annotates as requiring reasoning — a stratum the models barely memorize (closed book 0.07–0.15) and where a pilot measurement showed the highest share of cross-extractor-verifiable facts (0.80 vs. 0.44 on the balanced sample) — at least one serialized-graph condition beats gold passages with a separated interval on every model: the FIBO graph wins by 0.060 on gpt-oss, the schema-free graph by 0.055 on MiniMax, and on DeepSeek — the model that loses with graphs on every other stratum — *both* graph conditions win, by 0.085 and 0.131, with provenance pointers its best condition (all intervals exclude zero; Table 1). A prospective prediction registered before this experiment ran — that each model's grounded-correct lift (the number of grounded-correct answers gained over passages, of 169 numeric-gold cases) would exceed its balanced-sample value — landed two of three — gpt-oss and DeepSeek rose, MiniMax fell against a ceiling with passages already grounding 148 of 169 cases — and is reported as failed under its own all-models clause. The diagnostic's directional signal (Section 4) coincided with the reversal on all three models; its registered predictive form did not survive its first prospective test.

Table 1: Primary metric (share of the gold answer's numbers the answer states) by question stratum; "+ptr" adds provenance pointers to the FIBO graph. Bold marks graph conditions that beat passages with a 95% paired bootstrap interval excluding zero; the reasoning stratum is the only one where that happens, and there it happens on every model.

Stratum	Model	closed book	passages	graph (no schema)	graph (FIBO)	graph (FIBO+ptr)
balanced	gpt-oss	.306	.398	.355	.384	.364
	MiniMax	.185	.484	.490	.447	.502
	DeepSeek	.058	.295	.221	.233	.283
arithmetic	gpt-oss	.119	.530	.346	.374	.367
	MiniMax	.143	.553	.451	.474	.470
	DeepSeek	.054	.309	.182	.226	.271
reasoning	gpt-oss	.153	.665	.709	**.725**	.707
	MiniMax	.072	.763	**.819**	.817	.852
	DeepSeek	.026	.467	**.548**	**.574**	.634

Table 2: Grounded-correct counts on the balanced sample (199 numeric-gold cases): answers that were correct *and* whose serving evidence contained the gold numbers. Bold marks the best condition per model.

	passages	graph (no schema)	graph (FIBO)	graph (FIBO+ptr)
gpt-oss-120b	107	111	105	112
MiniMax-M2.7	124	138	127	**146**
DeepSeek-V3.1	98	83	82	104

4 Implications

The opening problem was that answers drawn from LLM-built graphs must be verifiable, not merely plausible. The findings convert into three deployment rules that serve exactly that requirement. First, align and deduplicate extracted facts by their position in the source document, not by entity name: every value conflict that name matching hid in this study was a unit-scale error that would otherwise reach an answer unexplained. Second, apply the ontology after extraction rather than inside the prompt: as guidance it made extractors disagree more, while as a validator it turned hundreds of silent defects per condition into checkable violations. Third, score any retrieval comparison evidence-conditionally: on public corpora, a scoring rule that cannot tell evidence use from memory rewarded the wrong condition on two of three models. Each rule's precondition is measurable from one pilot extraction (Appendix F). The graph earns its place in a regulated pipeline not by answering better but by making answers auditable — which is what the opening problem asked for.

5 Limitations

Three families of limitation bound our claims. *Scope*: one corpus and domain, three extractor models; passages are a retrieval ceiling, not a retriever; the graph is tested as serialized context, not as a queryable store. *Mechanism*: all models are accessed through APIs, so internals such as attention are unobservable — the model-dependent effects we report (over-refusal on serialized graphs, the pointer response) are characterized behaviorally, and their causes remain open. *Instruments*: scoring is anchored to gold answers, "evidence contains the answer" is a numeric-token approximation, strata below n=20 are underpowered, and the LLM-judge panel — calibrated, cross-judging, never load-bearing — inherits the known limits of LLM judging; its operational record is in Appendix E.

6 Conclusion

Ontology guidance, measured end to end, provides detectability rather than extraction quality; the alignment primitive that decides what "the same fact" means should be provenance, not names; and evaluation that cannot see evidence cannot see what structure is for. The value of LLM extraction is not the quality of its triples but their auditability — and auditability comes from provenance, not from the ontology. Three registered follow-ups remain unrun: extraction-contract alignment slots, an annotated-passages control for the pointer effect, and predictor validation of the diagnostic.

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GenAI Usage Statement

In this work we used a generative AI coding and research assistant (an LLM-based agent) for core research content: implementing the experiment harnesses and analysis code, refining author-proposed hypotheses into registered form, drafting substantial parts of this manuscript, and executing the pre-registered analyses. All direction — the research questions, the study design choices, the decision to pre-register, and every scope decision — came from the authors, and we have reviewed all AI-assisted work. Every load-bearing number in this paper is machine-verified against frozen run artifacts by a checker included in the supplementary material; the assistant's code was exercised end-to-end by the reported runs; and all registered verdicts, including the failed ones, are reported as measured. Separately from writing assistance, LLMs are also this study's *subjects*: the extractor and answering models named in the paper are experimental conditions, not tools of manuscript preparation. We take responsibility for the final content of this work, including text, claims, and artifacts produced with the aid of generative AI.

A Related work (extended)

LLM knowledge-graph construction and schema guidance. Pipelines from GraphRAG [2] to LightRAG [3] and HippoRAG [6] extract entities and relations with an LLM and retrieve over the result; most accept a schema or entity-type list as guidance, and domain deployments reach for standards such as FIBO [1]. What that guidance buys has mostly been assumed rather than isolated: comparisons vary schema together with prompts, models, or chunking. We hold everything but the schema fixed, at two sample scales, and measure agreement, coverage, and violation detectability separately — which is what lets the benefit relocate from generation to validation.

Evaluating graph-augmented retrieval. Head-to-head evaluations report win-rates by LLM judge [2] or accuracy on multi-hop suites [4], and disagree with each other by corpus and setup. Two blind spots recur: node identity across systems is resolved by name, and correctness is scored against gold answers that public-corpus models partly memorize. Both blind spots are our objects of study rather than our instruments — alignment is keyed on source provenance, and every verdict is conditioned on what the served evidence could support.

Contamination-aware evaluation. That benchmark answers leak through pretraining is well documented [7], and closed-book baselines are the standard control. Our closed-book gates span 0.06–0.31 across models on the same questions, and the representativeness check shows a gate that passes on a balanced sample can fail on the corpus's natural mix — so we treat contamination not as a dataset property but as a per-model, per-stratum quantity, and score answers by evidence-conditional decomposition rather than by gold match alone.

Judging and finance-QA metrics. Finance QA scores what matters in this domain — the numbers — with execution accuracy under tolerance [8, 9]; general RAG evaluation leans on LLM judges and semantic similarity [10–12], whose position, verbosity, and self-preference biases are documented [13, 14]. Our primary metric is the finance-QA family adapted to prose gold answers; the judge panel is reference-anchored, cross-judging (no model grades its own answers), and calibrated on must-fail controls before its verdicts count.

B Study design: full detail

B.1 Dataset and sampling

FinDER pairs terse analyst questions with gold answers and the filing passages that support them (5,703 cases, eight categories). Its own annotations partition the corpus into three strata: *simple* questions (neither annotation), *arithmetic* questions (an operation label such as Division), and *reasoning* questions (flagged as requiring composition, no arithmetic). We run three samples that together cover the grid: a category-balanced 280-case sample (35 per category, seed 42, multi-reference preferred), a 140-case arithmetic-stratum sample (type-proportional), and a 204-case census of the reasoning stratum — a census rather than a sample, so there is no draw to dispute. The balanced sample contains zero arithmetic cases — the multi-reference preference excluded them, since 881 of 883 arithmetic questions carry one reference — which is itself a finding about sample design and the reason the supplement exists.

B.2 The schema spectrum

Five extraction conditions differ only in the schema handed to the extractor: **A** a schema-free floor (one class, one relation); **B** a hand-written 20-class schema; **C** 70 classes derived from FIBO, corpus-scoped; **D** = C plus FIBO synonyms and abbreviations; **E** = C plus the subsumption hierarchy. Everything else — documents, chunking, extractor models (DeepSeek-V3.1, gpt-oss-120b, MiniMax-M2.7), prompts, property floor, graph store — is held fixed, so schema is the only moving part. Conditions A and C are re-run at 280 cases; each case's graph is extracted independently per model, giving three *views* per case and condition.

Each condition exists to isolate one belief (Table 3). Condition C's classes are not hand-picked: every FIBO class is ranked by how many corpus documents mention its label (word-boundary matching, alias-aware so that abbreviation-only classes are not silently discarded), and the top 70 across five FIBO domains are kept — the ontology as the corpus actually uses it, not as its authors organized it. The conditions nest ($C \subset D$, $C \subset E$) so each contrast is attributable to one addition, and the manipulation is verified from the recorded prompts themselves: base system-prompt sizes measure

4.6K (A), 15.4K (B), 19.8K (C), 20.5K (D), 20.5K (E) characters — D's synonyms add only 724 characters to C, which sharpens the dose–response reading: a very small amount of non-canonical vocabulary produces a large fragmentation effect.

Table 3: The schema spectrum: what each condition supplies and what belief it isolates.

	The extractor receives	The belief it tests
A	one class, one relation	floor: what agreement costs nothing
B	20 hand-written financial classes	a small bespoke schema suffices
C	70 corpus-scoped FIBO classes	the standard ontology, as used
D	C + FIBO synonyms/abbreviations	richer vocabulary converges naming
E	C + the subsumption hierarchy	structural context improves typing

B.3 Pre-registration

Each run is governed by a registration committed before it starts, stating direction and the outcome that would disconfirm it; exploratory findings are labelled and their registered replications named. The hypothesis ledger and all registrations ship in the supplementary material. We report every registered hypothesis whose experiment ran, including all eleven that failed.

C The schema spectrum: full results

C.1 Name agreement falls as the schema grows

Cross-model agreement on entity names is highest under the schema-free floor and falls under every ontology condition (paired A−C difference in per-case comparable-name rate: $+0.069$, 95% CI $[+0.046, +0.091]$, 280 cases). The mechanism is structural: one declared class gives two extractors nothing to disagree about; seventy classes are seventy naming choices.

C.2 Content coverage does not rise

The registered hypothesis that FIBO buys content at the price of names — supported at 16 cases — *reversed* at 280: the FIBO condition covers fewer gold-answer numbers than A (-0.036 $[-0.067, -0.008]$). The ontology has no measured extraction benefit on this corpus.

C.3 What survives: detectability

Schema violations (missing required properties, undeclared labels, datatype breaks) number 529–1,038 per ontology condition and are mechanically checkable against the declared schema; the schema-free floor offers almost nothing to check. The ontology's contribution relocates from generation quality to *verification surface* — consistent with keeping heavy ontology reasoning out of the extraction path and using it as an offline validator.

C.4 Why the synonym condition could not matter

Only 30 of 5,703 questions (0.5%) require vocabulary mapping at all, and the abbreviations questions actually use (`rev`, `mgmt`) are analyst shorthand, not FIBO designations. Condition D was measuring a need the corpus does not have — a caution for scoping ontology vocabulary work by measured query demand rather than by ontology completeness.

The five-point dose–response at scale

The full schema spectrum at 280 cases (name comparable rate, case bootstrap, same cases and models throughout):

All three registered directions for this curve failed: agreement is not monotone in class count (B's twenty bespoke classes fragment naming below FIBO's seventy), and both synonym lists and hierarchy text separate *below* plain FIBO at scale after being indistinguishable at 16 cases. What predicts fragmentation is not how many classes are declared but how much non-canonical lexical material the prompt carries.

Table 4: The five-condition dose–response at 280 cases: cross-model name-agreement rate (comparable-key rate) with case-bootstrap 95% intervals. Agreement is not monotone in class count — the hand-written 20-class schema sits below full FIBO — and both synonym and hierarchy additions separate below plain FIBO.

	no schema	FIBO (70)	FIBO+synonyms	FIBO+hierarchy	hand-written (20)
rate	.293	.211	.140	.126	.118
95% CI	[.269,.319]	[.191,.232]	[.120,.163]	[.106,.146]	[.097,.142]

D Provenance-anchored alignment: full results

Two views of the same case hold "the same fact" if we can align their facts. The standard key is the entity name. We instead locate each extracted value at the unique numeric token it came from in the source passage (searching all unit scales, so a mis-scaled extraction still anchors), and align facts that anchor to the same token.

At 280 cases on the schema-free condition, name keys yield 992 comparable fact pairs with 324 disagreements; anchor keys yield 1,656 pairs with 606 disagreements, of which **485 are invisible to any name key** — the views named the fact differently *and* disagreed on its value. On the FIBO condition the multiplier is larger (717 vs. 1,983 pairs; 246 vs. 874 disagreements; 734 name-blind). Both replications were registered in advance. The disagreements the anchor exposes are dominated by unit-scale misreadings: 26.7% of anchored values match their source only after rescaling by 10^3 or 10^6.

Two consequences follow. For pipeline builders: entity resolution keyed on names silently discards most of the verification signal that running a second extractor was supposed to buy. For evaluators: cross-model verification has a hard ceiling regardless of key — 87% of facts are held by exactly one view, a share that *rises* with sample size (77.1% at 16 cases) — and agreement-gating trades four correct answers for every wrong answer it blocks (precision $0.951 \rightarrow 1.000$, recall $0.951 \rightarrow 0.764$; the registered hypothesis that gating helps was disconfirmed on its own criterion).

E Answering under five evidence conditions: full results

E.1 Design

Each question is answered under five evidence conditions — closed book, gold passages, serialized schema-free graph, serialized FIBO graph, and the FIBO graph with provenance pointers — by the same three models at temperature 0. The serializer is one code path; document-hierarchy nodes are excluded because they carry the passages verbatim. Passages are the *ceiling* of retrieval (the gold evidence in full), and the graph conditions serve the union of all three extractor views, the ceiling of what extraction captured. Primary metric: scale-normalized overlap between the gold answer's numbers and the produced answer's (the finance-QA standard family [8, 9]); a three-judge cross-judging panel (no judge grades its own model's answers) covers what number matching cannot.

Judge panel, operational record. All three judges pass the four must-fail calibration controls (identity, swap, scale corruption, refusal), and only after two prompt repairs the controls themselves forced; each answer is graded by the two judges that did not write it. Two-judge agreement reaches 0.79 (pairwise 0.76–0.80); ties stay unresolved; 86 of 1,800 gradings failed at the API and are counted rather than imputed. The panel's pattern matches the primary metric's — graph-condition answers are rarely judged flat-wrong (4–10 per condition against 37 closed-book) and abstain more often. Full verdict tables ship with the supplementary material.

E.2 The gate: contamination is real and stratified

Closed-book performance spans 0.058 (DeepSeek) to 0.306 (gpt-oss) on the balanced sample — public filings are memorized to very different degrees — yet passages beat closed book for every model (paired differences $+0.092$ to $+0.299$, all separated). On the arithmetic stratum the gate widens sharply ($+0.41$ on both models run): *computed* values cannot be recalled, making arithmetic questions the contamination-resistant stratum for evaluating retrieval on public corpora.

E.3 Naive scoring: text wins or ties; format effects are model properties

Under naive overlap no graph condition beats passages anywhere; DeepSeek loses with graphs on both ($+0.075$, $+0.062$ toward passages), gpt-oss on the schema-free graph, MiniMax ties. The passages$-$graph gap itself differs across models beyond resampling (DeepSeek vs. MiniMax: $+0.081$ $[+0.019, +0.144]$) — averaging models erases a real interaction, which is one reason published RAG-vs-GraphRAG aggregates disagree.

E.4 Evidence-conditional evaluation reverses the reading

Scoring against the gold answer alone cannot distinguish a model that used the evidence from one that remembered the filing. Because we can check mechanically whether the served evidence contains the gold numbers, every verdict decomposes into: grounded-correct, utilization failure, over-refusal, honest abstention, contaminated-correct, and hallucination.

Table 5: The full evidence-conditional decomposition on the balanced sample (199 numeric-gold cases): every answer classified by whether the served evidence contained the gold numbers and what the model did. "Grounded" and "over-refusal" condition on evidence that contains the answer; "honest abstention," "contaminated," and "hallucination" on evidence that does not.

Model	Condition	grounded	util. fail	over-refusal	honest abst.	contam.	halluc.
gpt-oss	passages	107	17	31	9	22	12
gpt-oss	graph (no schema)	111	9	35	25	7	11
gpt-oss	graph (FIBO)	105	9	32	19	15	18
gpt-oss	graph (FIBO+ptr)	112	10	36	15	10	15
MiniMax	passages	124	10	21	15	22	6
MiniMax	graph (no schema)	138	6	11	26	10	7
MiniMax	graph (FIBO)	127	5	14	27	18	7
MiniMax	graph (FIBO+ptr)	146	3	9	17	11	12
DeepSeek	passages	98	26	31	11	20	12
DeepSeek	graph (no schema)	83	17	55	22	5	16
DeepSeek	graph (FIBO)	82	11	53	24	13	15
DeepSeek	graph (FIBO+ptr)	104	22	32	14	10	16

Three facts move (Table 5). The graphs' grounded yield matches or beats passages on two of three models. Passages carry 2–4× the contamination (20–22 vs. 5–10 answers scored correct on numbers absent from the evidence). And DeepSeek's graph deficit is over-refusal — 55 declines where the graph *contains* the answer, vs. 31 with passages — a legibility failure of the serialization for that model, not a failure of the graph's content. The registered replication on the arithmetic stratum did NOT reproduce the reversal: passages' grounded-correct count beats every graph condition on all three models there (109/121/96 vs. 80–106), the over-refusal mechanism does reproduce (50/41 graph declines vs. 24 with passages, repaired to 26 by pointers), and contamination is too rare on computed numbers to compare (3–11 per cell, pre-declared uninformative). The decomposition's reading is therefore stratum-dependent, which the main text carries as a finding (Section 3).

E.5 Provenance pointers raise accuracy without carrying content

Attaching each anchored value's source pointer (passage index and character offset — never the source text) was registered to change attribution, not accuracy. The registration was disconfirmed in the interesting direction: accuracy *rose* on MiniMax ($+0.055$ $[+0.012, +0.097]$) and DeepSeek ($+0.050$ $[+0.022, +0.080]$). Since no answer content was attached, the pointer acts as a trust signal — it marks exactly the numbers that anchored to source — and the pointer condition shows the highest evidence grounding rate (73%/70% of answered numbers present in the served evidence). Mechanical citation checking lands at 42–58% verified, so pointer-following is far from solved; but provenance decoration is a zero-content prompt intervention that improves accuracy.

E.6 The arithmetic stratum: the original hypothesis meets its test

The graph-favorable hypothesis this study inherited — that structured extraction wins precisely on multi-step, compositional questions — was registered verbatim for the arithmetic supplement, with the honest prior against it. It failed decisively: gold passages beat every graph condition on every model, all intervals separated ($+0.079$ to $+0.184$). More telling, the evidence-conditional reversal above does *not* replicate here: passages' grounded-correct count beats every graph on all three models (109/121/96 vs. 80–106). On questions that require computing a value from several printed ones, the serialized graph genuinely under-serves even after memorization is stripped — plausibly because the computation needs co-located operands that extraction fragments across nodes. Contamination itself nearly vanishes on this stratum (3–11 answers per cell against 20–22 on the balanced sample), confirming arithmetic questions as the contamination-resistant slice: closed book collapses to 0.05–0.14 while the gate opens four times wider ($+0.41$).

Two effects did carry over. The over-refusal mechanism reproduces (DeepSeek declines 50/41 graph-condition cases it could have answered, vs. 24 with passages), and the provenance-pointer repair concentrates exactly there: pointers cut its over-refusals to 26 and raise its grounded count from 64 to 82 ($+0.045$ $[+0.013, +0.078]$). The pointer effect that the balanced sample located on MiniMax did not reproduce on MiniMax here — the anchors hypothesis, as registered, is disconfirmed in its model-specific form, and what survives is weaker and more honest: pointer-only provenance helps, but which model it helps depends on the stratum.

Together the two strata answer the deployment question differently, which is the finding: *whether a graph is worth serving is a property of the question stratum and the model, not of the representation alone* — and it is measurable in advance, which is what the pre-deployment diagnostic below is for.

F The verifiability profile and design rules

Owning the full chain — source token, extracted node, served evidence, answer, verdict — lets each failure class be traced to its cause and converted to a rule. Unit-scale misreads (26.7%) argue for anchor-verifying values at write time. Name fragmentation growing with schema size argues for provenance keys in entity resolution and for deploying ontologies as validators rather than extraction guides. The 87% single-view ceiling caps what multi-extractor redundancy can verify — the budget belongs in provenance. Model-specific over-refusal on serialized graphs makes serialization a per-model design surface. And contamination that survives inside evidence conditions makes evidence-conditional evaluation, not closed-book gates alone, the honest default for public-corpus benchmarks.

F.1 A verifiability profile as a pre-deployment diagnostic

The design decisions this paper evaluates — multi-extractor verification, ontology guardrails, structural alignment — each pay off only under measurable preconditions, and every precondition is computable from one cheap pilot extraction, before any at-scale spend. We consolidate them: the *verifiable share* (facts held by ≥ 2 views: 13% here, so cross-view verification has a low ceiling); the *name-blind disagreement rate* (80–84%: how much signal a name-keyed pipeline discards); the *scale-fragility rate* (26.7%: the case for write-time anchor verification); the *guardrail surface* (violations per fact: the ontology's measurable value); per-model *legibility* (over-refusal on serialized evidence); and a structural component measured here for the first time. Anchors give cross-view node identity for free, so structural similarity of *the same* entity across views becomes well-defined: neighborhood agreement of anchor-identified entities is $4.7\times$ a matched null under the schema-free condition but only $2.3\times$ under FIBO — even when two views provably hold the same source fact, they embed it in almost disjoint structure (79% and 95% of same-fact pairs share no anchored neighbor at all), and the ontology fragments that structure further. A structural alignment method (SimRank-style or learned) has usable signal only where this ratio clears its null, which makes the ratio a go/no-go gate rather than a score. Validating the full profile as a predictor of downstream grounded-correct lift is deferred to follow-up work.

Model heterogeneity as a routing argument

Every component above varies across models given identical documents, schema, and prompts, and the variation is large enough to route on. The three extractors' facts overlap on only 13% of anchored values, so per-case extractor selection has measured headroom: an oracle choosing one model per case covers 0.458 of gold-answer numbers against 0.389 for the best fixed model (the oracle's picks split 103/49/47 across the three). Downstream, which evidence *format* a model can use is equally model-specific — the passages-minus-graph gap differs across models beyond resampling, one model over-refuses serialized graphs (55 vs. 24 declines) while another almost never does (2–5), and pointer decoration helps two models and not the third. A multi-agent system serving one representation to all models is therefore leaving measured accuracy on the table. The cautionary half of the argument is measured elsewhere: a learned router over partitioned stores can score below random selection when a miss serves no evidence at all, so the headroom quantified here should be harvested behind a fallback, not instead of one.

G Registrations and verdicts

Every registered hypothesis whose experiment ran, in plain language, with its verdict; registrations for follow-up experiments that have not yet run (annotated-passages, alignment-contract slots, pruned ontology, external corroboration) ship in the supplementary material without verdicts. The identifiers in the left column exist only to cross-reference the dated registration files in the supplementary material, each committed before the run it governs; nothing in the body text depends on them. Exploratory findings are absent from this table by construction — they are labelled in the text and their registered replications appear here.

H Scoring-metric sensitivity

The conclusions do not depend on the choice of primary metric, and the one comparison that would change is itself informative. Rescoring every balanced-sample answer with token-level F1 (the TAT-QA-style metric) preserves the gate on both models tested (passages − closed book $+0.071$ and $+0.082$, both separated) but flips two comparisons in directions the metric's own biases predict: passages separates above graphs everywhere (gold answers are written from the passages, so surface overlap is a gift to the passages condition), and the pointer condition scores *below* plain graph on gpt-oss (-0.016, separated) — because citation markers are tokens the gold answer does not contain, i.e., token F1 penalizes the act of citing. A metric that rewards passage-copying and punishes attribution cannot be primary in a study about verifiability; it is reported here so the reader can see exactly what it would have changed.

Two tokenizer defects were caught by a four-call smoke test before any scored run, and fixed symmetrically for gold and produced text: a hyphen after an alphanumeric read as a minus sign ("SP 800-171" yielding -171), and parenthesized enumeration markers "(1)...(6)" counted as numbers. The scoring tokenizer is separate from the anchoring tokenizer, which is left untouched because the anchors were located with it.

Family	Hypothesis (compressed)	Verdict
Second sweep (16 cases, five conditions)		
SW-H1	equal property floor removes period effect, not agreement	held
SW-H2	hierarchy (E) raises agreement over plain FIBO	not separated
SW-H3	FIBO content advantage survives equal slots	held at 16 (see S2-H4)
SW-H4	no ontology condition escapes name fragmentation	held
Scale-up, schema-free (280 cases)		
SC-H1	verifiable-fact shortage is sample-bound	held ($6\rightarrow 123$)
SC-H2	agreement-gated serving beats serve-always	**disconfirmed**
SC-H3	single-view ceiling does not move	held (rose to 87%)
SC-H4	anchor advantage survives scale (schema-free condition)	held
FIBO at scale (280 cases)		
S2-H1	anchor advantage replicates on the FIBO condition	held
S2-H2	name-agreement gap persists, paired	held
S2-H3	scale-error rate stable ($\approx$ a quarter)	held (26.7%)
S2-H4	FIBO content advantage survives scale	**disconfirmed, reversed**
Answering, balanced sample		
AN-H1	models cannot already answer (gate)	held 3/3
AN-H2	graph does not beat its source text	held
AN-H3	passages$-$graph gap is a model property	held (1/3 pairs separated)
AN-H4	anchors change attribution, not accuracy	**disconfirmed** on 2/3
Arithmetic supplement		
AR-H1	gate holds on arithmetic	held ($+0.41$)
AR-H2	a graph condition beats passages (the original claim)	**disconfirmed**
AR-H4	anchor effect reproduces on MiniMax	**disconfirmed**; moves to DeepSeek
AR-EC1	grounded-correct reversal replicates	**disconfirmed** (stratum-dependent)
AR-EC2	contamination asymmetry	uninformative (pre-declared)
AR-EC3	over-refusal concentrates on graphs (DeepSeek)	held
Schema dose–response at scale (280 cases)		
S5-H1	name agreement monotone in class count	**disconfirmed** (20-class lowest)
S5-H2	synonyms do not separate from plain FIBO	**disconfirmed** (separate below)
S5-H3	hierarchy does not separate from plain FIBO	**disconfirmed** (separate below)
Arithmetic supplement, grounding		
AR-H3	graphs ground more answers than passages	held (all models; .16 vs .48–.70)
Reasoning census (204 cases)		
RS-H1	gate holds on the reasoning stratum	held ($+0.44$ to $+0.69$)
RS-H2	passages $\geq$ both graph conditions	**disconfirmed**: graphs win, 3/3 models
RS-EC1–3	decomposition replicates on this stratum	held (contam. small: 3 vs 0–1)
PR-H1	pilot verifiable share predicts grounded lift	**disconfirmed** (2 of 3 models hit)
Representativeness (natural mix, graphless)		
RP-H1	gate replicates per model	partial: MiniMax yes, gpt-oss **no**
RP-H2	balanced design did not distort gate size	held (intervals overlap)
Multi-agent adjudication over anchored disagreements		
MA-H1/H3	provenance beats deliberation / pick-only rule	**void**: ground truth defective
	— all 1,482 disagreements are unit-scale splits (audited pre-scoring)	
MA-H2	schema does not resolve value conflicts	not separated (n=75; 71% abstain)